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Preprint · July 2024

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A Comparison of Delta and Rectangular Wings for Micro Aerial Vehicle Application

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Nomenclature

A = constant

C_D = drag coefficient

c_d = sectional drag coefficient

C_{Do} = zero lift drag coefficient

C_L = lift coefficient

C_R = root chord

K_P = potential constant

K_V = vortex constant

Λ = leading edge sweep angle

α = angle of attack

ε = semi-apex angle of wing

γ = constant

μ = viscosity

ρ = density

Subscripts

D = Delta

LE = leading edge

R = Rectangular

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Ref = reference

SE = side edge

min = minimum

max = maximum

I. Introduction

The low Reynolds number operation of micro aerial vehicles typically drives their configuration designs to low AR slender planforms. Consequently, numerous studies have quantified the performance of low aspect ratio rectangular, delta, circular, elliptic, Zimmerman, inverse Zimmerman, etc., geometries [1]-[8]. Of these configurations, the most copiously documented and simplest geometrically are thin planar delta [10]-[15] and rectangular wings [1]-[8]. Furthermore, studies do not indicate any significant benefit to the circular or Zimmerman type geometries compared to that of a rectangular wing [3]. A direct comparison of rectangular and delta wings however, does not appear to be documented.

Thin planar rectangular and delta wings have low sensitivity to Re such that variations in their operational Reynolds number do not yield significant discontinuities or deviations in their aerodynamic characteristics [9][10]; consequently trends established at low Re can be extrapolated to higher (nominally incompressible) Re and vice versa.

In terms of application, it would be of value to the designer to know if either a rectangular or delta wing configuration may offer aerodynamic advantage when compared with the constraint of equal planform area and aspect ratio; consequently yielding an analysis based on actual lifting ability in terms of force. As such, this article compares the performance of these geometries to establish if one is preferable to the other. Initially, existing methods to analytically analyze both wing forms are discussed and then compared with existing experimental data for validation. Results are then presented both based on analytic predictions as well as direct experimental comparison to assess comparative performance. Thus, the article aims to assess the query as to whether there is advantage to the use of either a rectangular or delta wing planform, and if so, under what conditions.

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II. Theoretical Modeling

2.1 Rectangular Wings

As shown in Ref. [9], a slender rectangular wing can be modeled using equations derived by Purvis [16] that define the coefficients in Polhamus' leading edge suction analogy [12] relation presented as Eq. (1). Lift is assumed to be composed of an attached flow component (i.e. a bound circulation based lift in the absence of leading edge suction) as well as vortex lift components generated along the leading and side edges.

$$C_L = K_{P\text{-}R}\cos(\alpha)^2\sin(\alpha) + \left(K_{V\text{-}LE} + K_{V\text{-}SE}\right)\cos(\alpha)\sin(\alpha)^2 \tag{1}$$

The attached flow potential constant as given by Larson [17] (which is essentially a curve fit to the vortex lattice results of Lamar [11][15]) is

$$K_{P\text{-}R} = \frac{2\pi AR}{\left[2+\sqrt{(4/3)AR^2+4}\right]} \tag{2}$$

Purvis gives an expression for the leading edge vortex lift as [16]

$$K_{V\text{-}LE} = \left(K_P\cos(\alpha)^2 - \frac{K_P^{\,2}}{\pi AR}\cos(\alpha)^5\right)/\cos(\Lambda) \tag{3}$$

while the side edge vortex suction is approximated by Purvis as [16]

$$K_{V\text{-}SE} = 4.91924\frac{K_P^{\,2}}{\pi AR^2}\cos(\alpha)^2 \tag{4}$$

As shown in Ref. [9], the inclusion of the leading edge suction term for rectangular wings under steady state conditions may not be representative of the flow physics. Consequently, the lift for a slender rectangular wing is well (and more consistently) approximated by inclusion of the attached flow potential lift and side edge vortex lift alone:

$$C_{L\text{-}R} = \frac{2\pi AR}{\left[2+\sqrt{(4/3)AR^2+4}\right]}\cos(\alpha)^2\sin(\alpha) + 4.91924\frac{K_P^{\,2}}{\pi AR^2}\cos(\alpha)^3\sin(\alpha)^2 \tag{5a}$$

substituting $K_{P\text{-}R}$ yields

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$$C_{L\text{-}R} = \frac{2\pi AR}{\left[2+\sqrt{(4/3)AR^2+4}\right]}\cos(\alpha)^2\sin(\alpha) + 4.91924\frac{\left(\frac{2\pi AR}{\left[2+\sqrt{(4/3)AR^2+4}\right]}\right)^2}{\pi AR^2}\cos(\alpha)^3\sin(\alpha)^2 \tag{5b}$$

2.2 Delta Wings

The lift coefficient for a delta wing (both non and slender) is well characterized by [13]

$$C_{L\text{-}D} = 2\pi\tan(\varepsilon)\left(1-\frac{2}{3}\sin(\varepsilon)\right)\sin(\alpha)\cos^2(\alpha) + \left(\frac{2\pi}{\cos(\varepsilon)}\left(1-\frac{2}{3}\sin(\varepsilon)\right)\left(\frac{1}{2}+\frac{1}{3}\sin(\varepsilon)\right)/\left(1+e^{\sigma(\Lambda_0-\Lambda)}\right)\right)\sin^2(\alpha)\cos(\alpha) \tag{6}$$

where (Λ_0 = 60 deg and σ = 0.23). In Eq. (6), the attached flow potential constant is given by

$$K_{P\text{-}D} = 2\pi\tan(\varepsilon)\left(1-\frac{2}{3}\sin(\varepsilon)\right) \tag{7}$$

III. Results and Discussion

Figure 1 shows a comparison of the theoretical lift curve slope (K_P) for a delta wing, rectangular wing and that from slender wing theory [= (π/2)AR [18]]. For an AR < 0.75, all three estimates show concurrence, while for AR > 0.75 both equations (2) and (7) show a steady reduction in the rate of increase of the lift curve slope with aspect ratio. The lift curve slope of the rectangular wing is consistently greater than that of the delta wing for the same AR, with a lift curve slope approximately 12% higher for the AR range presented.

Figure 2 shows comparisons of experimental data for delta wings with leading edge sweep angles ranging from 50 to 80 deg with predictions from Eq. (6). As seen, This relation accurately captures the functional dependency of the generated lift coefficient on wing AR. Predictive capability of Fig. (5b) for thin sharp rectangular wings is shown in Fig. 3. Agreement between theory and experiment is seen to be excellent when compared to experimental data extracted from various sources for thin rectangular wings. A notable difference observable in the experimental data (compared to Fig. 2 for delta wings) is the appearance of an apparent stall for the α range presented. For the delta wings (see Fig. 2), and irrespective of the wing's sweep, stall may be present but is not visually impactful on the lift curves, it appears more as a gentle rounding of the lift curve. The rectangular wings

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with AR ≥ 1.5 show a more pronounced stall, see Fig. 3, with a marked plateau in the lift curve post C_{Lmax}. Increasing AR shows an earlier appearance/onset of stall.

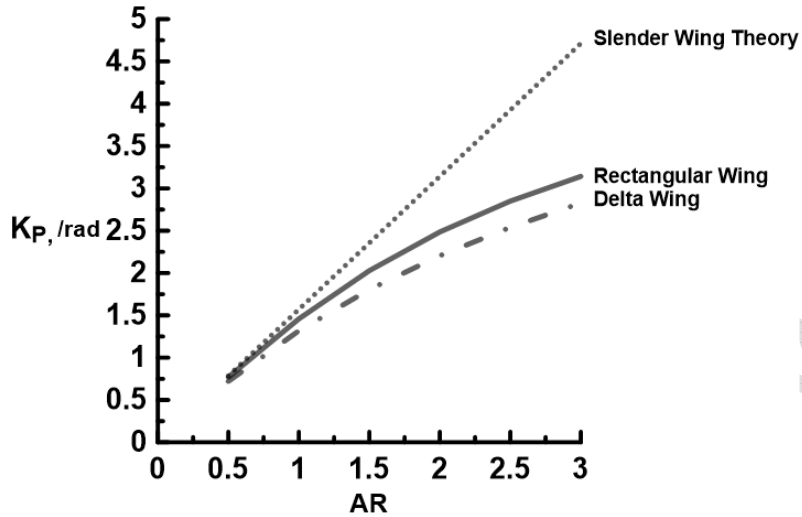

Fig. 1 Effect of aspect ratio on the attached flow lift curve slope of rectangular and delta wings.

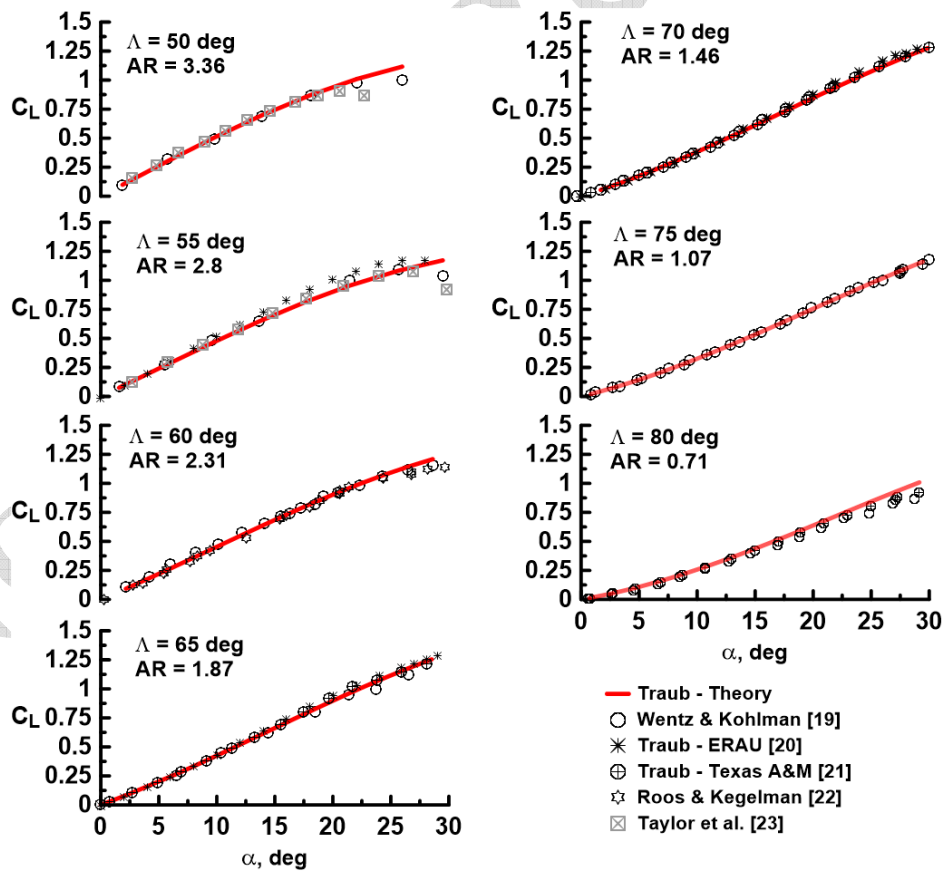

Fig. 2 Comparison of Eq. (6) with experimental delta wing data.

As Eqs. (5) and (6) show close accord with experiment they may be used to compare lifting performance of delta and rectangular wings, as presented in Fig. 4. For an AR < 2, lifting performance of a delta and rectangular wing is comparable. However, for an AR > 2, the slender rectangular wing shows superior lifting performance to that of an equal AR delta wing. The performance disparity increases with AR such that a non-slender delta wing appears as a poor lifting choice for wing's with AR > 2. A limitation to interpretation of the data in Fig. 4 is the lack of stall onset in the rectangular wing data. Consequently, Fig. 5 addresses this issue by direct comparison of experimental data for both wing geometries with similar ARs. As the delta wing ARs do not exactly match those of the rectangular wings in Fig. 3, their lifting performance was adjusted to compensate using Eq. (6). As such, the experimental C_L data values were scaled by the ratio of Eq. (6) C_L predictions for the actual experimental AR and that required or vice versa. Results for AR = 1, 1.5 and 3 are presented in Fig. 5. A higher AR is seen to increase lifting performance of the rectangular wing compared to the delta planform; however, increasing AR is also noted to cause earlier stall onset for the rectangular geometry with concomitant performance degradation. This performance compromise is clearly evident for AR = 3, the superior lifting performance at low to moderate angle of attack for the rectangular planform is offset by stall greatly reducing performance at higher angles of attack compared to the delta wing geometry.

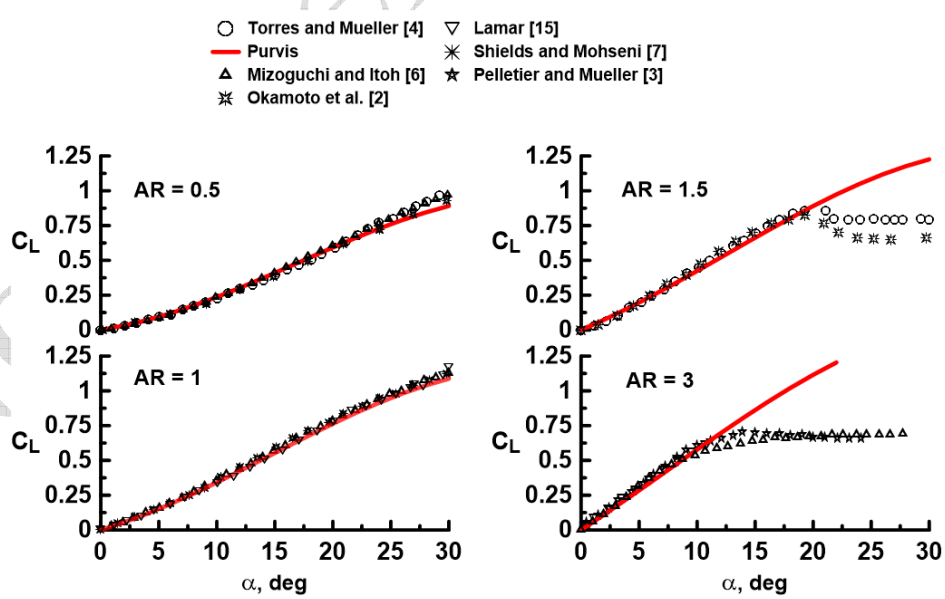

Fig. 3 Comparison of Eq. (5) with experimental rectangular wing data.

The overarching result from Figs. 4 and 5 is that for slender equal AR (< 2) rectangular and delta wings, lifting performance is similar pre-stall. Increasing the AR of the wings shows a (primarily) attached flow lifting benefit for the rectangular wing that is rapidly nullified by far inferior stall and post stall performance. Figure 4 and 5 ostensibly imply that a delta wing geometry is preferable to that of a rectangular wing when performance is considered over a large angle of attack range. While that may be so in terms of C_L, it remains to be assessed if this is true when viewed from the perspective of aerodynamic efficiency.

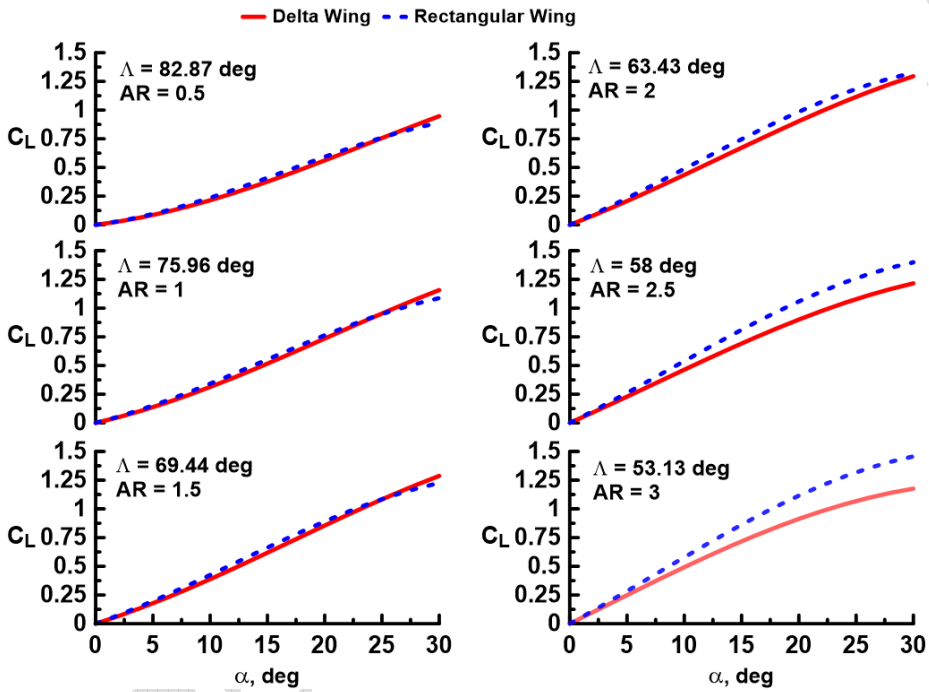

Fig. 4 Theoretical comparison of delta and rectangular wing lifting performance.

3.1 C_{Do} Considerations

For a thin planar wing, the drag due to lift is simply given by $C_L\tan(\alpha)$. The aerodynamic benefit of either a delta or rectangular geometry may be evaluated through examination of the lift to drag ratio. While most of the experimental data in Figs. (2) and (3) had associated drag plots, differences in model specifics, tunnel flow quality, surface finish etc, would contaminate the use of these experimental C_{Do} values to estimate C_L/C_D. A less subjective approach is to establish corresponding values of C_{Do} based on the assumption of fully laminar and turbulent flow, essentially setting a lower and upper bar for zero lift drag estimates.

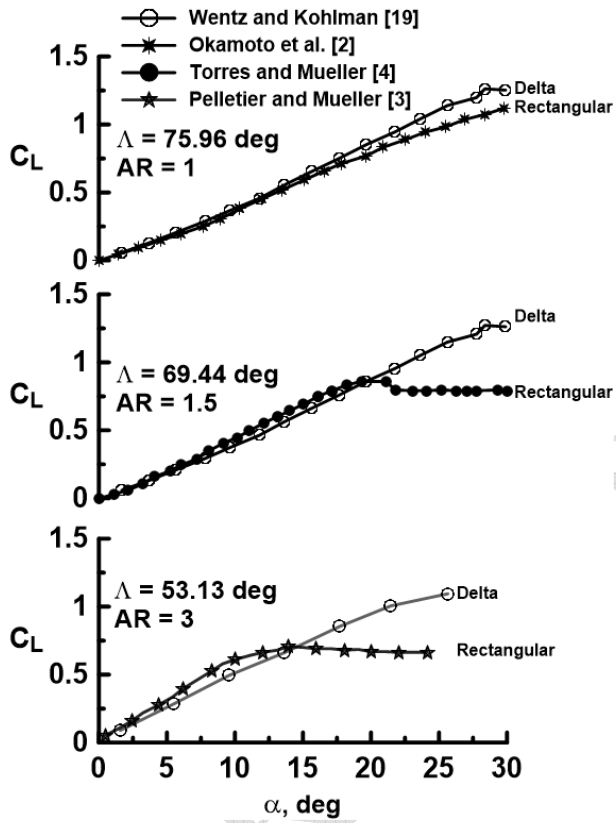

Fig. 5 Experimental lift performance comparison for delta and rectangular wings.

To make the comparison more meaningful, a suitable frame of reference is to compare a delta and rectangular wing with the same aspect ratio and surface area. In this instance, an equal C_L or C_D value implies an equal lift or drag force. For a constrained planform area and aspect ratio, it can easily be shown that

$$C_{R\text{-}D} = 2C_{R\text{-}R} \tag{8}$$

Let the sectional zero lift drag coefficient per wing surface (i.e. top or bottom) be expressed as

$$c_d = A \times Re^{\gamma} \tag{9}$$

such that for laminar flow A = 1.33 and $\gamma = -1/2$ while for turbulent flow A = 0.074 and $\gamma = -1/5$ [24]. Thus, for the rectangular wing upper or lower surface

$$C_{Do\text{-}R} = A \times Re^{\gamma} = A\left(\frac{\rho U C_{R\text{-}R}}{\mu}\right)^{\gamma} \tag{10}$$

and for the delta wing (integrating from the root to tip) surface

$$C_{Do\text{-}D} = \frac{2A}{2+\gamma} \times Re^{\gamma} = \frac{2A}{2+\gamma}\left(\frac{\rho U C_{R\text{-}D}}{\mu}\right)^{\gamma} \tag{11}$$

C_{Do} for the wing is then twice that in Eq. (10) or (11). Dividing Eq. (11) by Eq. (10) and noting that $C_{R\text{-}D} = 2C_{R\text{-}R}$ yields

$$C_{Do\text{-}D}/C_{Do\text{-}R} = \frac{2^{(\gamma+1)}}{2+\gamma} \tag{12}$$

Assuming fully laminar ($\gamma = -1/2$) or fully turbulent flow ($\gamma = -1/5$) shows that an equal planform area and aspect ratio delta wing will always have a slightly lower zero lift drag coefficient than a corresponding rectangular wing ($C_{Do\text{-}D}/C_{Do\text{-}R} = 0.94$ – fully laminar, and 0.97 – fully turbulent).

The experimental lift data in Fig. 5 was coupled with theoretical values of C_{Do} as given by Eqs. (10) and (11) allowing formulation of C_L/C_D [$= C_L/(C_{Do}+C_L\tan(\alpha))$] plots under the assumption of fully laminar (Re evaluated at 10^5) or fully turbulent (Re set to 10^6) flow. Results are presented in Fig. 6, with peak lift to drag ratios summarized in Table 1. Wing efficiency is seen to be somewhat comparable ($(C_L/C_D)_{max}$ is slightly higher for the rectangular wing) for delta or rectangular wings for AR $\leq$ 1.5 irrespective of whether the boundary layer is laminar or turbulent; however, the earlier stall onset of the rectangular wing gives it a smaller α envelope in which to operate efficiently. Increasing the AR of the wings to 3 shows a marked performance benefit for the rectangular wing, a consequence of its improved lifting performance yielding a lower drag due to lift. The overarching result from Fig. 6 is that for slender delta or rectangular wings (AR $\leq$ 1.5), efficiency is comparable, however, higher AR's favor a rectangular geometry although this must be offset against a notably reduced stall angle.

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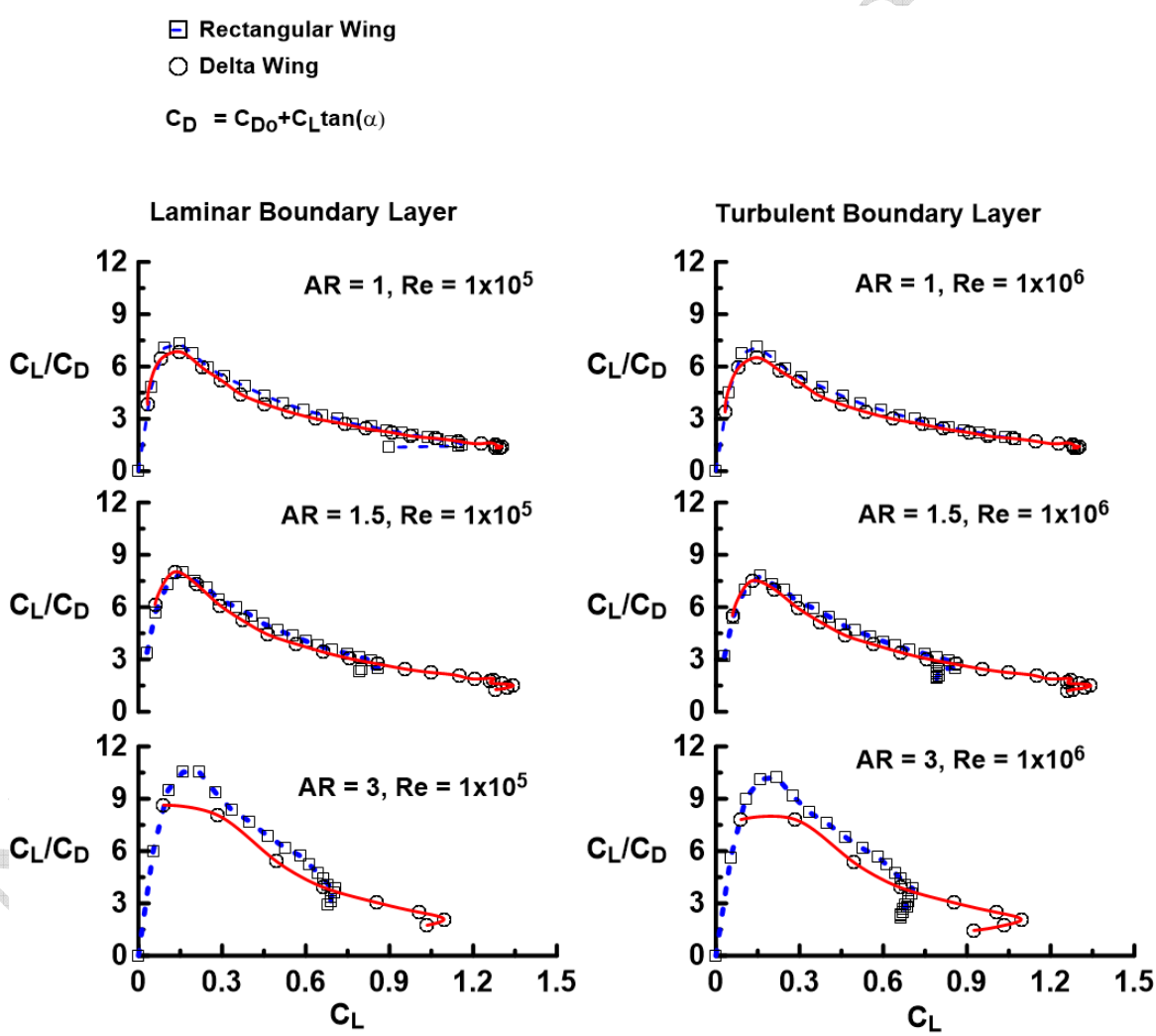
Table 1 Comparison Of Maximum Lift To Drag Ratio For Delta And Rectangular Wings

AR	$((C_L/C_D)_{max}$ Laminar Re = 100,000		$((C_L/C_D)_{max}$ Turbulent Re = 1,000,000	
	Delta	Rectangular	Delta	Rectangular
1	6.85	7.34	6.52	7.12
1.5	8.03	8.03	7.52	7.79
3	8.64	10.54	7.81	10.23



a) b)

Fig. 6 Efficiency comparison for delta and rectangular wings assuming a) laminar and b) turbulent boundary layer.

Conclusions

The performance of thin planar low to moderate aspect ratio rectangular and delta wings is examined under the constraint of equal AR and planform area. Analysis contains both experimental and theoretical elements. The results show that slender (AR ≤ 1.5) delta and rectangular wings have similar lifting performance and wing efficiency. The zero lift drag coefficient for the delta wing is always slightly lower than that of the rectangular wing. For higher aspect ratio's, the rectangular planform shows superior lifting performance yielding higher lift to drag ratios than the delta planform. However, these benefits need to be contrasted against a significant reduction in the stall angle giving the rectangular geometry wing a reduced flight envelope compared to that of a delta wing.

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